



THE SCOTS COLLEGE

Mathematics Extension 1

Trial Examination

9th August 2011

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A page of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Begin a new booklet for each question

Total Marks – 84

- Attempt Questions 1-7
- All questions are of equal value

Question 1 (12 marks) Start a SEPARATE writing booklet

a) Find $\int \frac{1}{\sqrt{4-x^2}} dx$ (1)

b) Sketch the region of the plane defined by $y \geq |3x - 2|$ (2)

c) State the domain and range of $y = \sin^{-1}(x^2)$ (2)

d) Using the substitution $u = 2x^5 - 1$. Find $\int x^4 (2x^5 - 1)^3 dx$ (3)

e) The point P(3,6) divides the line segment joining A(1,2) and B(x,y) internally in the ratio 2:1. Find the coordinates of B. (2)

f) The acute angle between the lines $y = 2x - 3$ and $y = mx + 1$ is 30° . Find the two possible values of m . (2)

Question 2 (12 marks) *Start a SEPARATE writing booklet*

- a) Find $\frac{d}{dx}(3\sin^{-1}4x)$. (2)
- b) A particle moves on the x -axis with velocity v . The particle is initially at rest at $x = 2$. Its acceleration is given by $\ddot{x} = x + 2$. Using the fact that $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$, find the speed of the particle at $x = 4$. (3)
- c) (i) Differentiate $e^{2x}(\sin x + 2\cos x)$ (2)
- (ii) Hence or otherwise, find $\int e^{2x}\cos x \, dx$ (1)
- d) A curve has parametric equation $x = 2t, y = 2t^2$. Find the Cartesian equation for this curve (2)
- e) Evaluate $\sum_{n=6}^8 (3n - 1)$ (1)
- f) Evaluate the limit of $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$ (1)

Question 3 (12 marks) Start a SEPARATE writing booklet

a) (i) Show that $e^x = \sin x + 3$ has a root between $x = 1$ and $x = 2$. (1)

(ii) Starting with $x = 1.5$, use one application of Newton's method to find a better approximation for this root. Write your answer correct to three significant figures. (2)

b) A particle moves in a straight line and its position at time t is given by

$$x = 3\cos\left(2t + \frac{\pi}{4}\right)$$

(i) Show that the particle is undergoing simple harmonic motion. (2)

(ii) Find the amplitude of the motion. (1)

(iii) When does the particle first reach its maximum speed after time $t = 0$? (1)

c) (i) Starting from the identity $\sin(\theta + 2\theta) = \sin\theta\cos 2\theta + \cos\theta\sin 2\theta$, and using the double angle formulae, prove the identity

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta \quad (2)$$

(ii) Hence find the general solution for the equation

$$\sin 3\theta = \sin\theta \quad (3)$$

Question 4 (12 marks) Start a SEPARATE writing booklet

a) The polynomial $P(x) = 2x^3 - 5x^2 + kx + 40$ has roots α, β, γ .

(i) Find the value of $\alpha + \beta + \gamma$. (1)

(ii) Find the value of $\alpha\beta\gamma$. (1)

(iii) It is known that two of the roots are equal in magnitude but opposite in sign. (2)
Find the third root and hence find the value of k .

b) Solve $\frac{5x}{x-2} \leq 3$ (3)

c) A grain silo dispenses grain at a constant rate of $7m^3$ per minute. As the grain falls it forms a cone shape such that the height of the cone is twice its radius.

(i) Show that $\frac{dV}{dh} = \frac{\pi}{4}h^2$ (2)

(ii) Find the rate at which the height is changing when its height is $2m$. (3)

Question 5 (12 marks) Start a SEPARATE writing booklet

a) $\int_0^{\frac{\pi}{4}} \sin^2(4x) \, dx$ (3)

b) (i) Sketch the curve $y = 3\cos^{-1}\left(\frac{x}{2}\right)$ for $-2 \leq x \leq 2$. (2)

(ii) On the same set of axes, sketch $y = 2x^2 - 2$. (1)

(iii) State how many roots the equation $3\cos^{-1}\left(\frac{x}{2}\right) - 2x^2 + 2 = 0$ has in the domain $-2 \leq x \leq 2$. (1)

c) Consider the function $f(x) = 4\tan^{-1}x$.

(i) State the range of the function $y = f(x)$. (1)

(ii) Sketch the graph of $y = f(x)$. (2)

(iii) Find the equation of the tangent to the curve $y = f(x)$ at $x = \sqrt{3}$. (2)

Question 6 (12 marks) *Start a SEPARATE writing booklet*

- a) Wilhemy's Law states that the rate of transformation of a substance in a chemical reaction is related to its concentration according to;

$$\frac{dE}{dt} = k(E - c)$$

where E is the amount of substance transformed, and c is the initial concentration of the substance.

- (i) By integration show that $E = Ae^{kt} + c$ is a solution to the given rate of change, where A is a constant. (2)

- (ii) Initially none of the substance is transformed. If the initial concentration is 8.3 and the amount transformed after 3 minutes is 2.9. Find how much of the substance will be transformed after 5 minutes, to 2 significant figures. (2)

- b) Consider the curve, $f(x) = 3 + \frac{1}{2x-5}$, $x \neq \frac{5}{2}$

The region enclosed by the curve of $f(x)$, the x -axis, and the lines $x = 3$ and $x = a$, is revolved through 360° about the x -axis. Let V be the volume of the solid formed.

Given that $V = \pi \left(\frac{28}{3} + 3\ln 3 \right)$, find the value of a . (3)

- c) A cylinder has height H and radius R . Point X is at one end of the cylinder, on the bottom and on the circumference. Point Y is directly opposite, halfway up the cylinder. Length $XY = D$.

- (i) Show that the volume of the cylinder is given by

$$V = \frac{\pi H}{16} (4D^2 - H^2) \quad (2)$$

- (ii) Find the maximum volume of the cylinder in terms of D if D is fixed. (3)

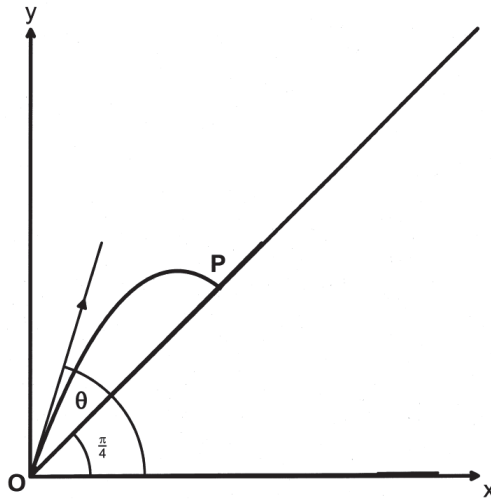
Question 7 (12 marks) *Start a SEPARATE writing booklet*

- a) Use mathematical induction to prove that

(4)

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{(n+1)^2}\right) = \frac{n+2}{2n+2} \text{ for all } n \geq 1.$$

- b) A tennis ball is hit with a velocity of 10m/s . Initially it is at O . P lies on an inclined plane. The inclined plane OP makes an angle of $\frac{\pi}{4}$ to the horizon.



The tennis ball is projected at an angle of θ to the horizontal and $\frac{\pi}{4} < \theta < \frac{\pi}{2}$.

The tennis ball has position at any time t given by

$$x = 10t\cos\theta \quad \text{and} \quad y = -5t^2 + 10t\sin\theta \quad (\text{Do not derive these equations})$$

- (i) If $OP = R$ meters and the tennis ball lands at P . Show that

$$x = y = \frac{R}{\sqrt{2}} \quad (1)$$

- (ii) Show that

$$R = 20\sqrt{2}(\cos\theta\sin\theta - \cos^2\theta) \quad (3)$$

- (iii) Find the maximum value of R .

(4)

END OF EXAM

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE : $\ln x \equiv \log_e x, \quad x > 0$

SOLUTIONS

ANSWER BOOKLET



Name: _____

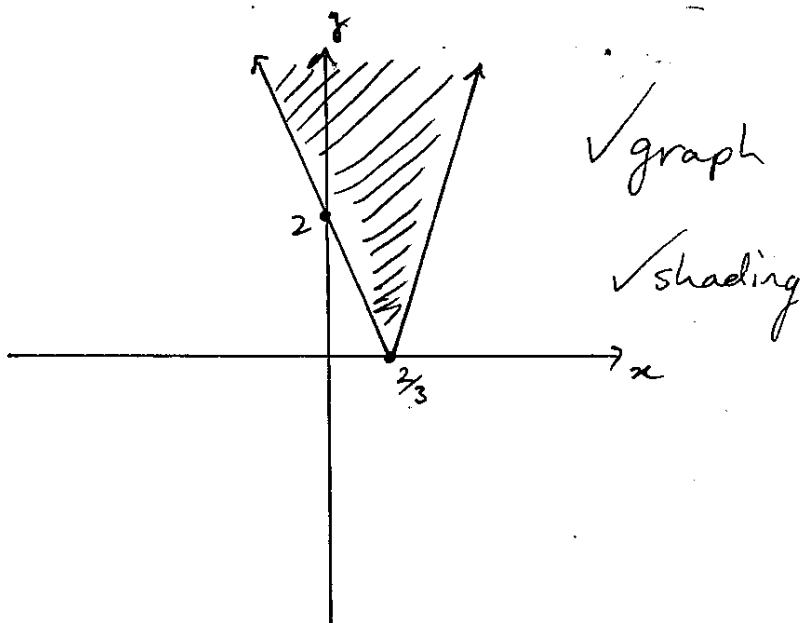
Teacher: _____

Question No. ①

a) $\int \frac{1}{\sqrt{4-x^2}} dx = \sin^{-1}\left(\frac{x}{2}\right) + C$ ✓ solution

b) $y \geq |3x - 2|$

test (0,0)
 $0 \geq |0 - 2|$
 $0 \geq 2$ Not true



c) Note:
 $y = \sin^{-1}(x)$

 $y = \sin^{-1}(x^2)$

$\therefore y = \sin^{-1}(x^2)$

domain: $-1 \leq x \leq 1$ ✓

Range: $0 \leq y \leq \frac{\pi}{2}$ ✓



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (1)

$$u = 2x^5 - 1$$

$$d) \int x^4 (2x^5 - 1)^3 dx$$

$$\frac{du}{dx} = 10x^4$$

$$du = 10x^4 dx$$

$$\frac{1}{10} \int 10x^4 (2x^5 - 1)^3 dx$$

/correct substitution

$$\frac{1}{10} \int u^3 du$$

$$\frac{1}{10} \left[\frac{u^4}{4} + c \right]$$

/integral

$$\frac{u^4}{40} + c$$

$$\frac{(2x^5 - 1)^4}{40} + c$$

/solution

Note: +C not
put down
loss of 1 mark.



ANSWER BOOKLET

Name: _____

Teacher: _____

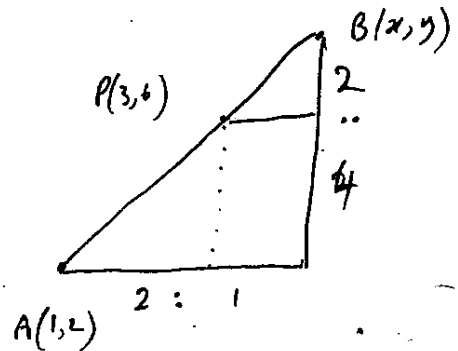
Question No. (1)

e) $P(3,6)$

$B(4,8)$

✓ correct method

✓ solution



f) $y = 2x - 3$, $y = mx + 1$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 30 = \left| \frac{2 - m}{1 + 2m} \right|$$

$\therefore$

$$-\frac{1}{\sqrt{3}} = \frac{2 - m}{1 + 2m}$$

$$m = \frac{-1 - 2\sqrt{3}}{2 - \sqrt{3}}$$

$$m = -5\sqrt{3} - 8$$

OR

$$\frac{1}{\sqrt{3}} = \frac{2 - m}{1 + 2m}$$

$$m = \frac{2\sqrt{3} - 1}{2 + \sqrt{3}}$$

$$m = 5\sqrt{3} - 8$$

✓ formula with values
two possible answers

✓ solution



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (2)

$$\frac{d}{dx} (3\sin^{-1}(4x))$$

let

$$y = 3\sin^{-1}(4x) \quad \text{find } \frac{dy}{dx}$$

$$\text{let } u = 4x \quad \frac{du}{dx} = 4$$

$$y = 3\sin^{-1}(u)$$

$$\frac{dy}{du} = \frac{3}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{12}{\sqrt{1-16x^2}} \quad \checkmark$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (2)

b) $t=0, x=2, v=0$

$$\ddot{x} = x + 2$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = x + 2$$

$$\frac{1}{2} v^2 = \frac{x^2}{2} + 2x + C \quad / \text{ using substitution \& finding } C.$$

when $x=2, v=0$

$$0 = \frac{4}{2} + 4 + C$$

$$0 = 6 + C$$

$$-6 = C$$

$\therefore$

$$\frac{1}{2} v^2 = \frac{x^2}{2} + 2x - 6$$

$\therefore$ at $x=4$

$$\frac{1}{2} v^2 = \frac{16}{2} + 8 - 6$$

$$\frac{1}{2} v^2 = 10$$

$$v = \pm \sqrt{20}$$

$\therefore v = \pm \sqrt{20}$ / solution
Speed is $\sqrt{20}$ when $x=4$.



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (2)

d) i) $e^{2x} (\sin x + 2 \cos x)$

 $u \times v$ ✓ attempts
product
rule

$$\frac{d}{dx} = 2e^{2x} (\sin x + 2 \cos x) + e^{2x} (\cos x - 2 \sin x)$$

$$= 4e^{2x} \cos x + e^{2x} \cos x$$

$$= 5e^{2x} \cos x \quad \text{✓ solution}$$

$$u = e^{2x} \quad u' = 2e^{2x}$$

$$v = \sin x \quad v' = \cos x - 2 \sin x$$

ii) $\int e^{2x} \cos x \, dx$

$$\frac{1}{5} \int 5e^{2x} \cos x \, dx$$

$$\frac{1}{5} [e^{2x} (\sin x + 2 \cos x)] + C$$

$$\frac{1}{5} e^{2x} (\sin x + 2 \cos x) + C$$

✓
solution



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (2)

d) $x = 2t$, $y = 2t^2$

$$\frac{x}{2} = t$$

$$\therefore y = 2\left(\frac{x}{2}\right)^2$$

$$= \frac{2x^2}{4}$$

$$y = \frac{x^2}{2}$$

/solving for t
/equating

e)
$$\sum_{n=6}^8 (3n-1) = [3 \times 6 - 1] + [3 \times 7 - 1] + [3 \times 8 - 1]$$
$$= 17 + 20 + 23$$
$$= 60$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (2)

f)

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin 2x}{2x}$$

$$\lim_{x \rightarrow 0} 2 \times \frac{\sin 2x}{2x}$$

$$\lim_{x \rightarrow 0} 2 \times \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$$

$$2 \times 1$$

$$2$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. 3

a) i) $e^x - \sin x - 3 = ?$

at $x=1$

$$e^1 - \sin(1) - 3 = -1.12$$

at $x=2$

$$e^2 - \sin(2) - 3 = 3.48$$

$\therefore$ change of sign there exists a root between $x=1$ & $x=2$.

ii) Newton's $\boxed{x = a - \frac{f(a)}{f'(a)}}$

$$f(x) = e^x - \sin x - 3$$

$$f(1.5) = 0.484$$

$$f'(1.5) = 4.41095$$

$$f'(x) = e^x - \cos x$$

at $a = 1.5$

$$x = 1.5 - \frac{0.484}{4.41095}$$

$$x = 1.5 - 0.1098$$

$$x = 1.39$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (3)

b i) $x = 3 \cos(2t + \frac{\pi}{4})$ where period is π
 $\therefore P = \frac{2\pi}{n}$
 $n = 2$

$$\dot{x} = -6 \sin(2t + \frac{\pi}{4})$$

$$\ddot{x} = -12 \cos(2t + \frac{\pi}{4})$$

$$\ddot{x} = -4 \times 3 \cos(2t + \frac{\pi}{4})$$

$$\ddot{x} = -4x$$

$\therefore$ this is in the form

$$\ddot{x} = -n^2 x$$

$\therefore$ particle is undergoing SHM.

b ii) amplitude is 3.

b iii) $\dot{x} = -6 \sin(2t + \frac{\pi}{4})$

$$-6 = -6 \sin(2t + \frac{\pi}{4})$$

$$1 = \sin(2t + \frac{\pi}{4})$$

$\therefore$

$$2t + \frac{\pi}{4} = \frac{\pi}{2}$$

$$t = \frac{\pi}{8}$$

$\therefore$ at time $\frac{\pi}{8}$ particle first reaches its max speed.



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. 3

$$\begin{aligned} \text{b i)} \quad \sin 3\theta &= \sin \theta \cos 2\theta + \cos \theta \sin 2\theta \\ &= \sin \theta [1 - 2\sin^2 \theta] + \cos \theta [2\sin \theta \cos \theta] \\ &= \sin \theta - 2\sin^3 \theta + 2\sin \theta \cos^2 \theta \\ &= \sin \theta - 2\sin^3 \theta + 2\sin \theta (1 - \sin^2 \theta) \\ &= \sin \theta - 2\sin^3 \theta + 2\sin \theta - 2\sin^3 \theta \\ &= 3\sin \theta - 4\sin^3 \theta \end{aligned}$$

$$\text{ii)} \quad \sin 3\theta = \sin \theta$$

$$3\sin \theta - 4\sin^3 \theta = \sin \theta$$

$$2\sin \theta = 4\sin^3 \theta, \quad \sin \theta = 0$$

$$\frac{2}{4} = \sin^2 \theta$$

$$\pm \frac{1}{\sqrt{2}} = \sin \theta$$

$\therefore$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4},$$

General solutions

$$\theta = \left(\frac{\pi}{4} \pm n\pi \right) \text{ or } \left(\frac{3\pi}{4} \pm n\pi \right) \quad \text{where } (n=0,1,2,\dots)$$

or $n\pi$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. (4)

$$P(x) = ax^3 + bx^2 + cx + d$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

a) i) $\alpha + \beta + \gamma = \frac{5}{2}$

ii) $\alpha\beta\gamma = \frac{-40}{2} = -20$

iii) let $\alpha = -\beta$

$\therefore -\alpha^2\gamma = -20 \dots (1)$

$\gamma = \frac{5}{2} \dots (2)$

$-\alpha^2\left(\frac{5}{2}\right) = -20$

$5\alpha^2 = 40$

$\alpha^2 = 8$

$\alpha = \pm\sqrt{8}$

$\therefore$

$\alpha = \sqrt{8}, \beta = -\sqrt{8}, \gamma = \frac{5}{2}$

$$\alpha\beta + \gamma\alpha + \beta\gamma = \frac{k}{2}$$

$$-8 + \sqrt{8}\left(\frac{5}{2}\right) - \sqrt{8}\left(\frac{5}{2}\right) = \frac{k}{2}$$

$$-8 = \frac{k}{2}$$

$$\underline{-16 = k}$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. 4

b) $\frac{5x}{x-2} \leq 3 \quad x \neq 2$

$$5x(x-2) \leq 3(x-2)^2$$

$$5x^2 - 10x \leq 3(x^2 - 4x + 4)$$

$$5x^2 - 10x \leq 3x^2 - 12x + 12$$

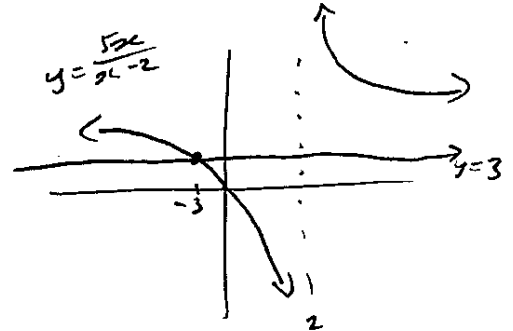
$$2x^2 + 2x - 12 \leq 0$$

$$2(x^2 + x - 6) \leq 0$$

$$2(x+3)(x-2) \leq 0$$

$\therefore$

$$\boxed{-3 \leq x < 2}$$





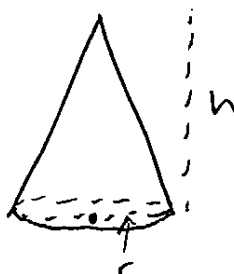
ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. 6

6)



$$2r = h$$

$$r = \frac{h}{2}$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi \left(\frac{h}{2}\right)^2 h$$

$$V = \frac{\pi}{12} h^3$$

$$\frac{dV}{dh} = \frac{3\pi}{12} h^2 = \frac{\pi}{4} h^2$$

$$ii) \quad \frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$$

$$= 7 \times \frac{4}{\pi h^2}$$

$$\frac{dh}{dt} = \frac{28}{\pi h^2}$$

when $h = 2$

$$\frac{dh}{dt} = \frac{28}{\pi \times 4}$$

$$= \frac{7}{\pi}$$

$$\frac{dV}{dt} = 7$$

$$\frac{dh}{dV} = \frac{4}{\pi h^2}$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. 5

$$a) \int_0^{\pi/4} \sin^2(4x) dx$$

$$\int_0^{\pi/4} \frac{1}{2} - \frac{1}{2} \cos(8x) dx$$

$$= \left[\frac{x}{2} - \frac{1}{16} \sin(8x) \right]_0^{\pi/4}$$

$$= \left[\frac{\pi}{8} - \frac{1}{16} \sin(2\pi) \right] - [0 - 0]$$

$$= \left[\frac{\pi}{8} - 0 \right] - 0$$

$$= \frac{\pi}{8}$$

$$\cos 2A = 1 - 2\sin^2 A$$

$$\sin^2 A = \frac{1}{2} - \frac{1}{2} \cos 2A$$



ANSWER BOOKLET

Name: _____

Teacher: _____

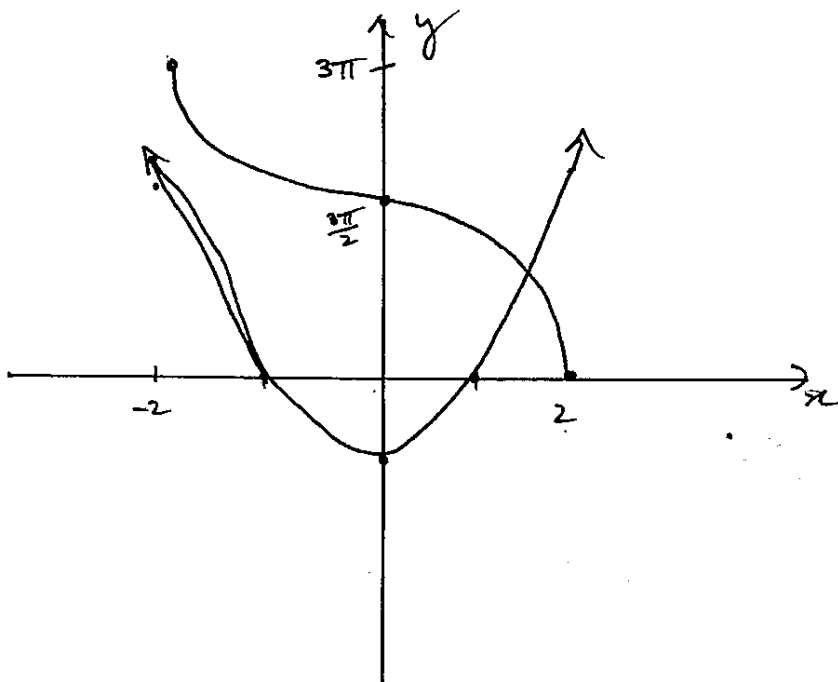
Question No. 5

b)

$$y = 3\cos^{-1}\left(\frac{x}{2}\right)$$

$$\frac{y}{3} = \cos^{-1}\left(\frac{x}{2}\right)$$

$$x = 2\cos\left(\frac{1}{3}y\right)$$



ii) $y = 2x^2 - 2$

at

x	-2	-1	0	1	2
y	6	0	-2	0	6

iii) on domain

$$-2 \leq x \leq 2$$

only 1 solution



ANSWER BOOKLET

Name: _____

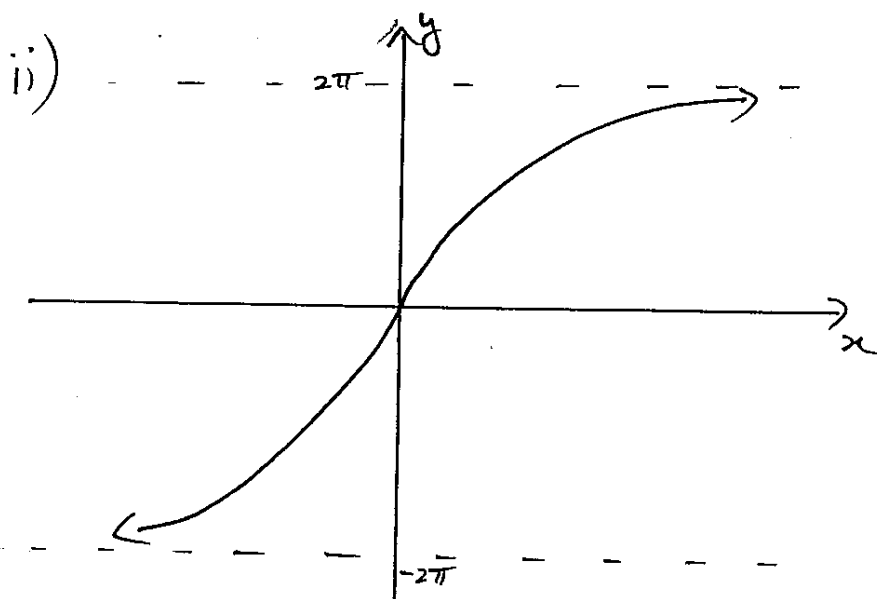
Teacher: _____

Question No. 5

c) i) range of $y = f(x)$

$$-4 \times \frac{\pi}{2} \leq y \leq 4 \times \frac{\pi}{2}$$

$$\boxed{-2\pi < y < 2\pi}$$



iii) $y = 4 \tan^{-1}(x)$

at $x = \sqrt{3}$

$$y = 4 \tan^{-1}(\sqrt{3})$$

$$\frac{dy}{dx} = \frac{4}{1+x^2}$$

$$= \frac{4\pi}{3}$$

at $x = \sqrt{3}$

Tangent

$$y - \frac{4\pi}{3} = 1(x - \sqrt{3})$$

$$\frac{dy}{dx} = \frac{4}{4} = 1$$

$$3x - 3y - 3\sqrt{3} + 4\pi = 0$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. 6

a) i $\frac{dE}{dt} = k(E-c)$

$$\frac{dt}{dE} = \frac{1}{k} \times \frac{1}{(E-c)}$$

$$t = \frac{1}{k} \int \frac{1}{E-c} dE$$

$$t = \frac{1}{k} \ln(E-c) + D$$

$$k(t-D) = \ln(E-c)$$

$$e^{k(t-D)} = E-c$$

$$\frac{e^{kt}}{e^D} = E-c \quad \left(\text{let } \frac{1}{e^D} = A\right)$$

$$Ae^{kt} = E-c$$

$$\therefore E = Ae^{kt} + c$$



ANSWER BOOKLET

Name: _____

Teacher: _____

Question No. 6

$$a) ii) \quad E = Ae^{kt} + c$$

$$\text{at } t=0 \quad E=0 \quad c=8.3$$

$$0 = Ae^0 + 8.3$$

$$A = -8.3$$

$$\therefore E = -8.3e^{kt} + 8.3$$

$$\text{at } t=3 \quad E=2.9 \quad c=8.3$$

$$2.9 = -8.3e^{3k} + 8.3$$

$$\frac{-5.4}{-8.3} = e^{3k}$$

$$\frac{54}{83} = e^{3k}$$

$$k = \frac{1}{3} \ln\left(\frac{54}{83}\right)$$

$$\therefore E = -8.3e^{kt} + 8.3$$

$$\text{at } t=5$$

$$E = -8.3e^{5k} + 8.3$$

$$= 4.245$$

$$= 4.2 \quad (2 \text{ sig fig})$$

Question No. 6

b)

$$V = \pi \int_3^a \left(3 + \frac{1}{2x-5}\right)^2 dx$$

$$= \pi \int_3^a \left(9 + \frac{6}{2x-5} + (2x-5)^{-2}\right) dx$$

$$= \pi \left[9x + 3 \ln(2x-5) + -\frac{1}{2}(2x-5)^{-1} \right]_3^a$$

$$= \pi \left[9x + 3 \ln(2x-5) - \frac{1}{4x-10} \right]_3^a$$

$$= \pi \left[\left(9a + 3 \ln(2a-5) - \frac{1}{4a-10}\right) - \left(27 + 0 - \frac{1}{2}\right) \right]$$

$$= \pi \left[9a - 27 + \frac{1}{2} - \frac{1}{4a-10} + 3 \ln(2a-5) \right]$$

$$\therefore \begin{aligned} 2a-5 &= 3 \\ a &= 4 \end{aligned}$$



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Question No. 6

b)

$\therefore$ if $a = 4$

$$9 \times 4 - 27 + \frac{1}{2} - \frac{1}{46-10} =$$

$$9\frac{1}{2} - \frac{1}{4} =$$

$$\frac{28}{3} = \text{as required.}$$



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Question No. 6

d) i)

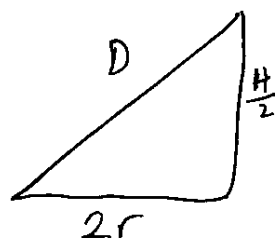
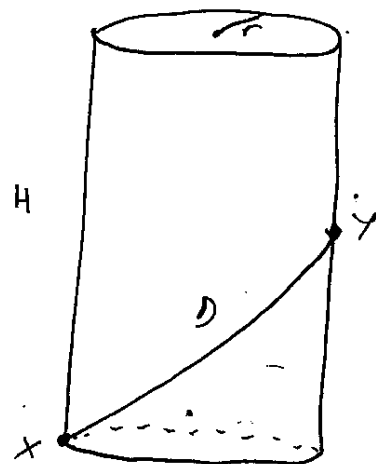
$$D^2 = \frac{H^2}{4} + 4r^2$$

$$r^2 = \frac{D^2}{4} - \frac{H^2}{16}$$

$$V = \pi r^2 H$$

$$V = \pi \left(\frac{D^2}{4} - \frac{H^2}{16} \right) H$$

$$V = \frac{\pi H}{16} (4D^2 - H^2)$$



ii)

$$V = -\frac{\pi}{16} H^3 + \frac{\pi D^2}{4} H$$

$$\frac{dV}{dH} = -\frac{3\pi}{16} H^2 + \frac{\pi D^2}{4}$$

at $\frac{dV}{dH} = 0$ is max or min

cubic is of form

$$y = -ax^3 + bx$$

 $\therefore$ take +ve solution

$$0 = -\frac{3\pi}{16} H^2 + \frac{\pi D^2}{4}$$

$$H^2 = \frac{4D^2}{3}$$

$$H = \pm \sqrt{\frac{4D^2}{3}}$$

only +ve solution

$$H = \frac{2D}{\sqrt{3}}$$



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Question No. 7a) Test for $n=1$

$$\left(1 - \frac{1}{2^2}\right) = \frac{1+2}{2+2}$$

$$\frac{3}{4} = \frac{3}{4}$$

 $\therefore$ True for $n=1$ ✓We assume it is true for $n=k$ ($k \in \mathbb{N}$)

i.e. ✓

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+2}{2k+2}$$

Now prove it is true for $n=k+1$

$$\underbrace{\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \dots \left(1 - \frac{1}{(k+1)^2}\right)}_{\text{from assumption}} \left(1 - \frac{1}{(k+2)^2}\right) = \frac{k+3}{2k+4}$$

from assumption

$$\frac{k+2}{2k+2}$$



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Question No. 7

a) continued

$$\left(\frac{k+2}{2k+2} \right) \left(1 - \frac{1}{(k+2)^2} \right) = \frac{k+3}{2k+4}$$

✓
progress

$$\frac{k+2}{2k+2} - \frac{k+2}{(2k+2)(k+2)^2}$$

$$\frac{(k+2)(k+2)^2 - (k+2)}{(2k+2)(k+2)^2}$$

$$\frac{[k+2][(k+2)^2 - 1]}{2(k+2)^2(k+1)}$$

$$\frac{(k+2)^2 - 1}{2(k+2)(k+1)}$$

$$\frac{k^2 + 4k + 4 - 1}{(2k+4)(k+1)}$$

$$\frac{(k+1)(k+3)}{(k+1)(2k+4)}$$

$$\frac{k+3}{2k+4} = R.H.S$$

 $\therefore$ true for $n=k+1$

$\therefore$ Because it is true
for $n=1$ it is true
for $n=2$ and by
mathematical induction
is true for $n=3, 4, \text{etc}$

✓ solution.



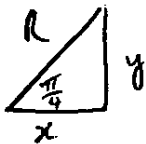
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Question No. (7)

b i)

isosceles
 $\therefore x = y$

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} = \frac{y}{R}$$

$$\frac{R}{\sqrt{2}} = y$$

 $\therefore$

$$x = y = \frac{R}{\sqrt{2}}$$

ii)

$$x = y$$

$$10t \cos \theta = -5t^2 + 10t \sin \theta$$

$$5t^2 - 10t \sin \theta + 10t \cos \theta = 0 \quad \checkmark \text{ setup}$$

$$5t(t - 2 \sin \theta + 2 \cos \theta) = 0$$

 $\therefore$

$$t = 0$$

or

$$t - 2 \sin \theta + 2 \cos \theta = 0$$

 $\therefore$

$$t = 2 \sin \theta - 2 \cos \theta \quad \checkmark \text{ solve for } t$$

 $\therefore$

$$x = 10t \cos \theta$$

$$\text{when } t = 2 \sin \theta - 2 \cos \theta$$

$$x = 10(2 \sin \theta - 2 \cos \theta) \cos \theta$$

but

$$x = \frac{R}{\sqrt{2}}$$

/ substitute in for x

$$\frac{R}{\sqrt{2}} = 20(\sin \theta \cos \theta - \cos^2 \theta)$$

$$R = 20\sqrt{2}(\cos \theta \sin \theta - \cos^2 \theta)$$

As required



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Question No. 7

iii)

$$R = 20\sqrt{2} (\cos\theta \sin\theta - \cos^2\theta)$$

$$\frac{dR}{d\theta} =$$

Product
RuleProduct
Rule

$$\frac{dR}{d\theta} = 0 \text{ for max.}$$

$$0 = 20\sqrt{2} (\cos 2\theta + \sin 2\theta) \quad \checkmark \frac{dR}{d\theta}$$

 $\therefore$

$$\cos 2\theta + \sin 2\theta = 0$$

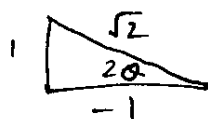
$$\sin 2\theta = -\cos 2\theta$$

$$\tan 2\theta = -1$$

2nd or 4th
but must be in
domain $\frac{\pi}{4} < \theta \leq \frac{\pi}{2}$

$$\therefore \text{2nd} \quad \frac{\pi}{2} < 2\theta < \pi$$

$$\begin{array}{c|c} S & A \\ \hline T & C \end{array}$$



$$\therefore \cos 2\theta = \frac{-1}{\sqrt{2}}$$

 $\checkmark$ solution

$$\sin 2\theta = \frac{1}{\sqrt{2}}$$

$$R = 20\sqrt{2} (\cos\theta \sin\theta - \cos^2\theta)$$

Note:

$$\cos\theta \sin\theta = \frac{1}{2} \sin 2\theta = \frac{1}{2\sqrt{2}}$$

$$\cos^2\theta = \frac{1}{2} - \frac{1}{2} \cos 2\theta = \frac{1}{2} - \frac{1}{2\sqrt{2}}$$

$$R = 20\sqrt{2} \left(\frac{1}{2\sqrt{2}} - \left(\frac{1}{2} - \frac{1}{2\sqrt{2}} \right) \right)$$

$$= 20\sqrt{2} \left(\frac{1}{2\sqrt{2}} - \frac{1}{2} + \frac{1}{2\sqrt{2}} \right)$$

$$= 20\sqrt{2} \left(\frac{1}{\sqrt{2}} - \frac{1}{2} \right)$$

$$= 20 - 10\sqrt{2}$$

 $\checkmark$ sub
in
valueMaximum value of $\checkmark$ exact
value.

$$R \text{ is } (20 - 10\sqrt{2})$$